From Raw Shots to Expectation Values: Shot Grouping Improves Time Series Prediction

Anonymous Authors

***Abstract*—Between keeping every shot as a noisy training example and averaging all shots into one clean feature lies an entire spectrum of data organizations that quantum machine learning usually leaves implicit. We make that choice explicit. Shot grouping partitions a fixed measurement record into disjoint groups and averages within each group, tuning smoothly between the two conventions with a single validated parameter and no additional quantum executions. We evaluate the method on chaotic synthetic benchmarks and on real-world data under strict execution budgets. An intermediate grouping consistently outperforms the standard expectation-value pipeline. Matched controls show that the gain acts as correctly scaled regularization of the classical readout. Unmitigated runs on two superconducting processors confirm the trend on every task. Shot grouping therefore offers a practical way to improve finite-shot quantum learning at no added quantum cost.**

***Index Terms*—quantum reservoir computing, finite shot learning, measurement aggregation, time series forecasting**

I. Introduction

Every quantum learning experiment ends the same way: a circuit runs many times, and each run returns one measured bitstring. Before anything can be learned, this pile of shots must become training data. The step looks like bookkeeping, and it is usually treated that way. Almost every pipeline averages all shots of a time step into a single expectation-value estimate and moves on. This paper asks what that habit costs.

The habit has a respectable origin. Expectation values are the native currency of quantum mechanics: they are what theory predicts and what experiments are built to estimate. A learning pipeline, however, does not consume observables. It consumes training examples, and it cares about two things only: how many there are and how noisy each one is. The moment a measurement record becomes training data, a physics convention quietly turns into a statistical decision: spend all measurements on making one example clean, or make several examples that are merely clean enough. No single choice can suit every task, budget, and noise level.

Averaging everything is one end of a dial. It buys the cleanest possible feature vector but pays with scarcity. Thousands of circuit executions collapse into a handful of training examples, and too few examples relative to the number of features destabilizes regression [1]. The opposite end, keeping every shot as its own example, buys abundance but pays with noise, and noisy features systematically shrink fitted coefficients [2]. Between these extremes lies an entire range of aggregations. Nothing in the record forces either convention, and the record does not care which one tradition prefers.

Quantum reservoir computing (QRC) is a clean place to study this dial [3]–[5]. A fixed quantum circuit transforms an input sequence into measured features, and only a classical readout is trained. The circuit, the targets, and the measurement record can therefore be held completely fixed while the aggregation alone changes. Whatever happens to the forecast is attributable to the aggregation because nothing else moved.

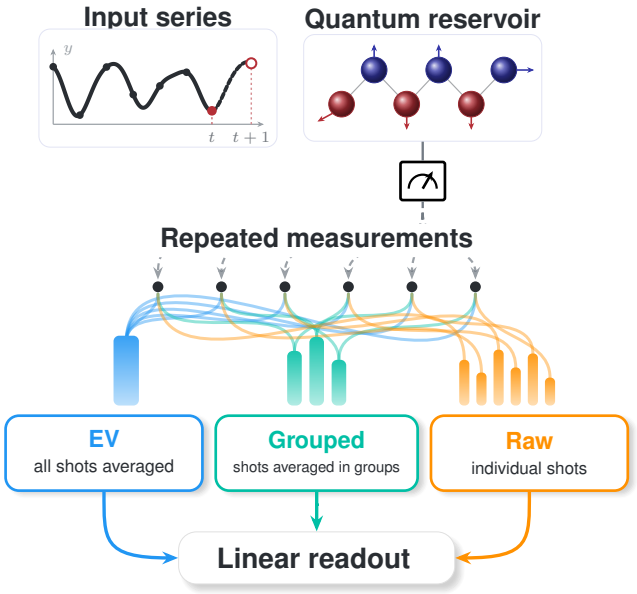


Fig. 1. Three aggregation modes for one finite measurement record. A normalized input drives a fixed quantum reservoir. The same repeated measurements are processed using full expectation-value averaging (EV), disjoint group averaging (Grouped), or individual shots (Raw). Only the classical aggregation of the measurement record changes.

We compare three representations of one record (Fig. 1). The expectation-value representation (EV) averages all measurements at each time step. Raw uses each measurement separately. Grouped averages disjoint subsets, and its aggregation scale k, the number of shots per averaged training view, slides smoothly from $k = 1$ (Raw) to $k = N$ (EV).

The dial turns out to matter. Under one matched protocol and fixed execution budgets, an intermediate aggregation lowers test error relative to full averaging on every task we evaluate, including two synthetic systems and two real-world datasets. The best aggregation scale depends on the record, so no single convention is appropriate. The gain requires nothing quantum. Matched controls trace it to correctly scaled regularization of the classical readout, supplied automatically by the record's own fluctuations. Because aggregation is classical post-processing, it costs no additional quantum executions and applies retroactively

to any archived record.

Our contribution is *adaptive shot aggregation*. The number of shots per training view becomes a setting of the classical readout, selected by validation like any other hyperparameter, at no additional quantum cost. We map when the choice matters through three budget allocations of time steps and shots. We explain why it works through matched controls that isolate the regularization mechanism. We test where it transfers through unmitigated experiments with a separate hardware reservoir on two superconducting processors.

II. Related Work

Measurement grouping has prior use. Cemin *et al.* average subsets of measurements to enlarge a training set for reduced quantum channel learning [6], and training with noisy inputs is related to an additional penalty on large fitted weights [7]. This study evaluates the complete range from Raw to EV on a fixed QRC record. It adds joint calibration of aggregation and readout regularization, an exact explanation for the resulting penalty, three allocations of a fixed execution budget, and a matched evaluation on four forecasting tasks. The contribution is the calibration and evaluation of the aggregation scale, not the original introduction of subset averaging.

Finite sampling has also been addressed at other stages. Ahmed *et al.* derive noise-aware training after expectation values have been formed [8]. Kreplin and Roth use variance regularization to reduce sampling noise [9], while resource-aware optimization allocates measurements before execution [10]. Sampling limits and useful combinations of features have also been studied in physical learning systems [11]. Our method begins after execution and uses a single archived measurement record.

QRC research has examined how noise, dissipation, measurement, finite Hilbert space, and input transformations shape temporal features [12]–[23]. Those studies analyze or modify the quantum representation. Here the representation is held fixed and only the organization of repeated measurements changes.

III. Method

A. Simulation reservoir

The reservoir has six qubits and two shallow data reuploading layers [24]. Each layer encodes the scalar input on every qubit with heterogeneous R_y and R_z rotations, applies neighboring R_{zz} couplings, and mixes locally with frozen non-Clifford angles. The circuit parameters are fixed before forecasting, and additional frozen realizations quantify sensitivity. One computational basis measurement yields six local Z_i signs and five neighboring products Z_iZ_{i+1} at no additional quantum cost.

Ideal statevector simulation is performed before forecasting. All 11 features change with the input and exceed the specified nonlinearity threshold. They all contribute distinct numerical information. Because several respond in similar ways, their effective number is 5.15. A leak of 0.2 and ten causal lags provide temporal context. Ridge regression produces the next step forecast, and performance is measured by test NRMSE. These checks show that the reservoir supplies a varied, nonlinear feature set. They do not evaluate quantum advantage.

B. Adaptive shot aggregation

Let $x_t^{(n)}$ denote the processed features from shot index n at time t. For a divisor k of N, the N shots are partitioned into $G = N/k$ disjoint groups, producing

$$x_t^{(g)} = \frac{1}{k}\sum_{n\in g} x_t^{(n)}, \qquad g = 1, \ldots, G. \tag{1}$$

Each grouped sample shares the same target. At evaluation, the G predictions for one step are averaged. This equals prediction from their mean features, so any difference between EV and Grouped is created during training rather than evaluation.

Measurement indices do not identify physical trajectories. A random permutation is therefore drawn at every time step before groups are formed, and the analysis is repeated across random grouping assignments. A fixed assignment across time is included as a sensitivity analysis.

C. Fixed execution budget and training

The quantum cost is TN circuit executions. Three specified settings use exactly $10{,}000$ executions. They allocate $(T, N) = (250, 40)$, $(500, 20)$, and $(1000, 10)$ to the number of time steps and shots. All divisors of N are considered, including Raw at $k = 1$ and EV at $k = N$. Three expanding train-and-check splits preserve time order: each candidate pair (k, λ) is fitted on an earlier block and scored on the following block. The best pair is then fixed, the readout is refitted to all data preceding the test block, and the final 20% is evaluated once. Normalization and feature scaling use training data only.

D. Why grouping regularizes the readout

Full averaging gives the readout one clean view of each time step. Individual shots give it many views, but every view is noisy. Intermediate groups provide several partially denoised versions of the same target. Differences among those versions reveal which feature changes are stable across shots and which are measurement noise. Training on all grouped versions discourages the readout from relying on the unstable changes.

For a linear readout with squared loss, this intuition can be written as an exact identity. Let $\bar{x}_t$ be the mean of the G grouped views and let $C_{k,t} = G^{-1}\sum_g (x_t^{(g)} - \bar{x}_t)(x_t^{(g)} - \bar{x}_t)^\top$. This quantity summarizes how the measured features fluctuate together around their mean. For intercept b and weights w,

$$\frac{1}{G}\sum_g \left(y_t - b - w^\top x_t^{(g)}\right)^2 = \left(y_t - b - w^\top \bar{x}_t\right)^2 + w^\top C_{k,t} w. \tag{2}$$

The first term on the right is the EV loss. The second adds a larger cost when the readout relies on combinations of features that vary strongly across the grouped views. Grouped training is therefore exactly EV training plus a penalty determined by the observed shot fluctuations. This identity requires neither Gaussian noise nor independence across time.

Cov-EV applies the same fluctuation-based penalty directly to EV and chooses its strength separately using the same earlier data as Grouped. If both methods perform similarly, the benefit of grouping is explained by this regularization rather than by additional physical information. We also compute both forms at fixed k and confirm identical predictions as an implementation check. The leaky integrator and overlapping lags make adjacent feature sets dependent, so uncertainty is computed over trajectories or measured windows, not over the many grouped views. The full derivation and statistical details are provided in the supplement.

IV. Experimental Setup

A. Simulation setup

We evaluate Mackey–Glass [25], Rössler [26], Santa Fe competition Series A far-infrared laser intensity [27], and monthly sunspots [28]. In simulation, all four tasks use the same frozen six-qubit reservoir, $Z+ZZ$ features, leak, ten lags, washout, training based scaling, time-ordered model selection, execution budgets, and candidate settings. The only task specific inputs are the source sequence and forecast window origins.

Each synthetic task contributes five independently generated trajectories. Each real-world dataset contributes five chronological windows with disjoint test blocks. Santa Fe uses five fully disjoint windows from the 9,093-point continuation record. Three circuit parameter settings, three measurement samples, and three grouping assignments repeat the experiment within each forecasting unit. We first average these repetitions within each unit and then compute paired 95% intervals across the five separate trajectories or windows. This prevents repeated simulations of one sequence from being counted as independent forecasting problems.

The main comparison includes the identically leaked and lagged scalar input, ideal reservoir features without sampling noise, EV, Grouped, Raw, and independently tuned Cov-EV training. Every sampled method reuses the identical raw record, time-ordered splits, ridge candidates, and test access policy. Twenty extra grouping assignments at the balanced setting test sensitivity to arbitrary group assignments.

After a washout of 30 steps, the last 20% of each usable sequence is reserved for testing and is not inspected while settings are chosen. The earlier block is evaluated in three rounds: each round fits on the past and checks the forecast on the next block. For every aggregation scale k, the ridge value is chosen from $\{10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^{2}, 10^{3}\}$; Cov-EV uses the same ridge values with eight predeclared strengths for the fluctuation-based penalty. Scaling is recomputed using only the fitting portion in every round. Scoring first averages all Grouped predictions back to one forecast per time step. Thus every method contributes the same number of forecasts to the comparison. The test block is accessed only after all choices have been fixed.

The allocation $(T, N) = (500, 20)$ is the primary comparison. The other two allocations use the same budget of 10,000 executions and are secondary analyses. Reservoir parameters, generators, window origins, repeat identifiers, grouping rules, candidate settings, and reporting filters are stored before the test blocks are evaluated. The reservoir, aggregation choices, ridge choices, and test procedure are identical across tasks.

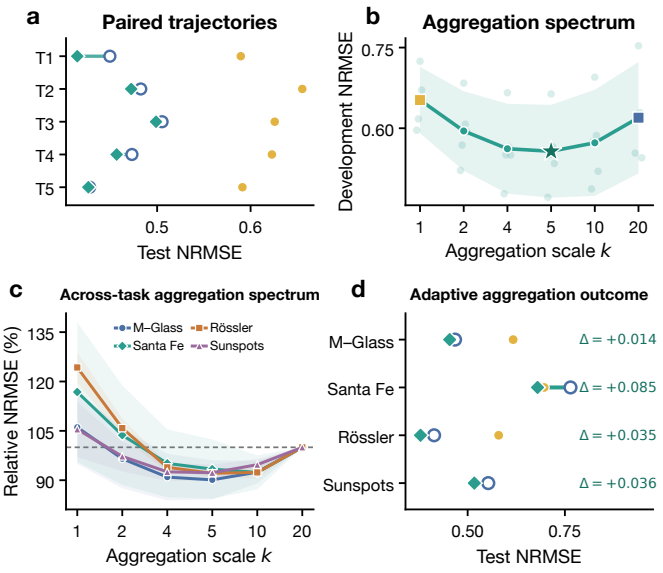


Fig. 2. Summary under the shared protocol. (a) Test outcomes for Mackey–Glass for five independent trajectories. (b) Balanced aggregation spectrum over aggregation scale. (c) The spectrum relative to EV across tasks. (d) Test outcomes after choosing aggregation scale and ridge strength together. Panels (b,c) use development data only; panels (a,d) use test blocks. Bands and intervals use separate trajectories or windows; faint marks show repeated measurement trials within them.

B. Hardware setup

The device study intentionally uses a different reservoir design. Its fixed ten-qubit circuit contains one amplitude-encoded input qubit, nine Hadamard-initialized memory qubits, one R_{ZZ} ring layer, transverse R_X rotations, and ten local-Z outputs. The circuit is mapped to selected physical qubits on *ibm_fez* and *ibm_marrakesh* and is held fixed across all four hardware tasks. It tests whether the same aggregation procedure transfers to a second circuit and set of measured features; it is not a hardware implementation of the six-qubit simulator or a qubit-count comparison. Within each study, every aggregation method receives the same circuit outputs and only the classical organization of shots changes. One $T = 500$, 400-shot acquisition per task and processor is partitioned into twenty ordered and disjoint records with $N = 20$. Santa Fe uses continuation window 5, $[4587, 5088)$, and was acquired later under the identical protocol. Hardware intervals describe variation among measurement blocks within one acquisition. They do not represent independent devices or calibration periods.

V. Evaluation

A. Balanced setting

Table I and Fig. 2 summarize the balanced setting. Grouped lowers mean NRMSE relative to EV by 0.0138 on Mackey–Glass (95% CI $[-0.0023, 0.0300]$), 0.0848 on Santa Fe laser ($[0.0174, 0.1522]$), 0.0349 on Rössler ($[0.0139, 0.0559]$), and

TABLE I

MEAN TEST NRMSE AT THE BALANCED SETTING. REPEATED MEASUREMENT TRIALS ARE AVERAGED WITHIN EACH OF FIVE SEPARATE TRAJECTORIES OR WINDOWS. LOWER VALUES ARE BETTER.

Task	Class.	Ideal	EV	Grouped	Raw	Cov-EV
Mackey–Glass	0.100	0.178	0.468	0.454	0.617	**0.447**
Santa Fe laser	0.762	0.493	0.764	0.680	0.697	**0.677**
Rössler	0.096	0.054	0.414	0.379	0.580	**0.367**
Sunspots	0.387	0.474	0.553	0.517	0.558	**0.515**

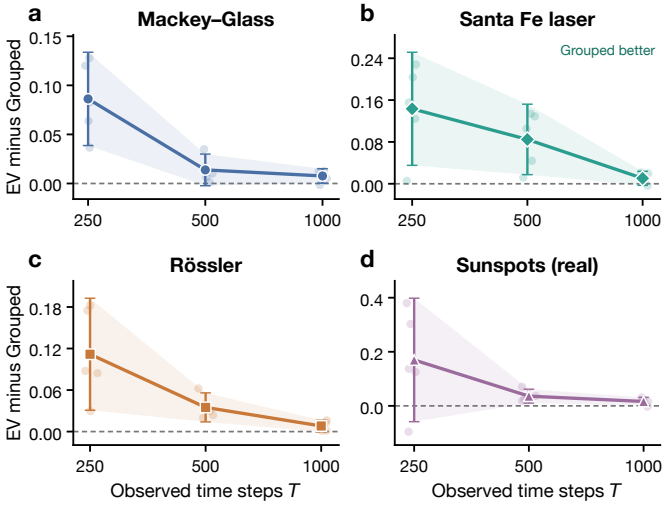


Fig. 3. Comparison at fixed execution budget. Each panel shows $\Delta = \mathrm{NRMSE}_{\mathrm{EV}} - \mathrm{NRMSE}_{\mathrm{Grouped}}$ at $(T, N) = (250, 40), (500, 20), (1000, 10)$. Positive values favor Grouped. Points are trajectory or window units; markers and bars are means and 95% intervals.

0.0360 on sunspots ($[0.0103, 0.0616]$). The intervals exclude zero for both real-world datasets and Rössler. The interval for Mackey–Glass includes zero.

Santa Fe has the largest difference at the balanced setting. Adaptive shot aggregation reduces EV from 0.764 to 0.680. Each of the five disjoint windows favors Grouped. Joint tuning favors $k = 10$ most often and an intermediate aggregation scale in 94% of repeated trials. Cov-EV reaches 0.677, nearly matching Grouped, while the Grouped gain decreases from 0.143 with 40 shots per step to 0.010 with 10 shots per step. This pattern is consistent with grouping changing the readout differently for different tasks and execution allocations.

The ideal QRC model performs worse than the identically processed classical input on Mackey–Glass and sunspots, better on Santa Fe, and yields an inconclusive difference on Rössler. The experiments therefore support claims about aggregation of a chosen reservoir record, but not quantum advantage.

The chosen aggregation scale varies across tasks, trajectories, circuit parameter settings, measurement samples, and grouping assignments. EV is preferred in some cases. Equation (2) shows that changing k changes which feature combinations the readout is discouraged from using, not only how strongly it is discouraged. Raw generally has higher error, while several intermediate aggregation scales remain competitive with EV.

B. Time steps and shots

Figure 3 compares three allocations of the same 10,000 executions. With 40 shots at each of 250 time steps, every task has a positive mean difference between 0.086 and 0.170. The intervals exclude zero for Mackey–Glass, Santa Fe, and Rössler, while the sunspot result varies across windows. With 10 shots at each of 1,000 time steps, the mean differences fall between 0.008 and 0.016. The intervals exclude zero for Mackey–Glass and sunspots and include zero for Santa Fe and Rössler. Grouping has its largest effect when the record contains many shots per time step and fewer time steps.

C. Controls and grouping robustness

Cov-EV has a lower mean than Grouped on all four tasks at the balanced setting. The paired differences range from 0.0021 to 0.0116 NRMSE, and only the Rössler interval excludes zero. Grouped obtains its fluctuation-based penalty directly from the record without introducing another adjustable strength. Cov-EV applies the same fluctuation-based penalty directly to EV and chooses its strength from earlier data. Their similar performance supports the explanation in Eq. (2).

Raw has higher error than EV on Mackey–Glass and Rössler at the balanced setting. The intervals for Santa Fe and sunspots include zero. The exact two-form calculation matches Grouped ridge at fixed k to numerical precision and verifies the implementation. It differs from Cov-EV, which chooses the strength of the fluctuation-based penalty independently. Across twenty repeated settings in the sensitivity analysis, EV minus Grouped means are -0.0064 on Mackey–Glass, 0.1103 on Santa Fe, 0.0177 on Rössler, and 0.0493 on sunspots; only the sunspot interval excludes zero. Fixed-group minus per-time-regrouped mean differences are at most 0.0021 NRMSE in magnitude and every clustered interval contains zero. The grouping convention therefore has little effect relative to variation across repeated settings. Figure 4 summarizes the matched controls for all four tasks.

D. Hardware transfer

All four tasks have positive mean EV minus Grouped differences on both processors (Fig. 5): 0.038–0.153 NRMSE on *ibm_fez* and 0.031–0.103 on *ibm_marrakesh*. For Santa Fe, EV decreases from 0.750 to 0.711 on Fez, a gain of 0.038 ($[0.027, 0.050]$), and from 0.744 to 0.713 on Marrakesh, a gain of 0.031 ($[0.023, 0.039]$). Grouped wins 19/20 blocks on each processor. Santa Fe was acquired later using the same circuit, physical layouts, and analysis. These paired intervals are descriptive of variation among measurement blocks within one acquisition. They do not measure variation across independent hardware campaigns.

VI. DISCUSSION

Adaptive shot aggregation is a small change to the classical side of a quantum learning pipeline. The complete measurement record is retained, and each aggregation scale k decides how many shots are combined into each training view. Earlier time blocks are used to choose k together with the ridge strength

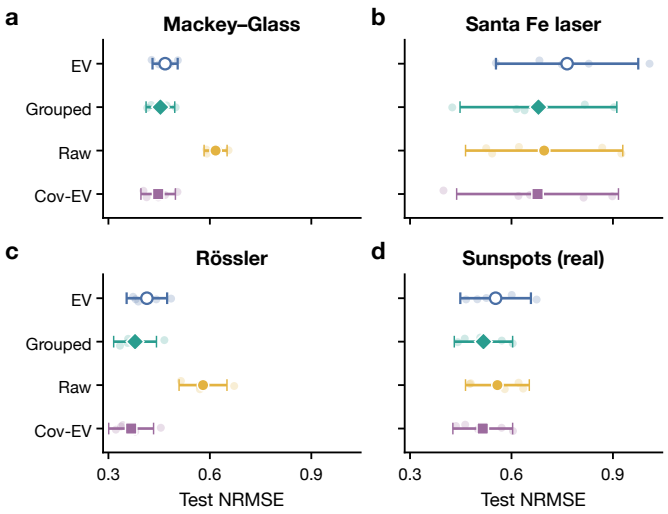


Fig. 4. Controls at the balanced setting in the order EV, Grouped, Raw, and Cov-EV. Faint points are the five trajectory or window means after repeated measurement trials are averaged within each unit; markers and bars show means and 95% intervals.

λ before the test block is opened. Raw ($k = 1$) and EV ($k = N$) are included as the two extremes, so either conventional representation is kept when partial averaging is not useful.

The practical advantage is that the method makes better use of measurements that have already been acquired. It requires no additional circuits, shots, mitigation, or trainable quantum parameters. Its largest gains appear when a record contains many shots per time step but relatively few time steps. In this regime, full averaging leaves only one training example for each time step, while individual shots are too noisy. Partial averaging provides several denoised views of each target and lets the readout adapt to the available record.

Equation (2) explains this behavior. Training on grouped views is mathematically identical to EV training with an added cost for relying on feature changes that vary strongly across shots. Cov-EV applies this cost directly and nearly matches Grouped. The gain therefore comes from stabilizing the classical readout, not from creating additional physical information.

The same pattern appears across synthetic dynamics, measured time series, and unmitigated hardware records. The feature audit verifies that the simulation reservoir responds to the input, provides nonlinear information, and contains no redundant measured feature. The separate ten-qubit hardware circuit then tests whether the aggregation procedure transfers to another reservoir design. It is not a qubit-count comparison, but the positive trend on both processors shows that the method is not tied to the six-qubit simulation circuit.

The method requires individual measurement outcomes because alternative aggregation scales cannot be reconstructed from stored expectation values. When bitstrings are retained, however, adaptive aggregation can be added after execution and evaluated at negligible cost. The present study covers two synthetic systems, two real-world datasets, linear readouts, and one acquisition per task and processor. Independent hardware campaigns and additional reservoir families are the natural next tests.

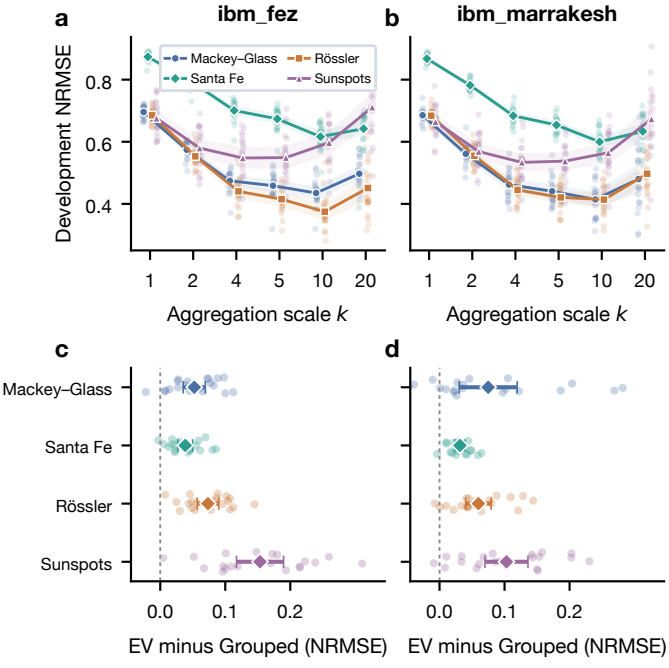


Fig. 5. Unmitigated transfer results from the separate ten-qubit reservoir on *ibm_fez* and *ibm_marrakesh*. For each task and processor, one record with $T = 500$ and 400 shots per step is partitioned into twenty disjoint records with $N = 20$. The upper panels show performance across aggregation scales. The lower panels show the test difference between EV and Grouped; positive values favor Grouped.

VII. Conclusion

Adaptive shot aggregation makes the representation of a finite measurement record a controllable part of the learning pipeline. Across all four forecasting tasks, intermediate aggregation reduced mean test error at the balanced execution budget. Unmitigated measurements on both quantum processors reproduced the positive trend. Equation (2) shows that grouped views stabilize the readout using the observed shot fluctuations while using exactly the same shots as Raw and EV. Retaining shot-level records therefore enables a direct post-execution improvement without changing the circuit or increasing the quantum execution budget.

Appendix A
Reproducibility Summary

All four simulation tasks use the same six-qubit reservoir, preprocessing, time-ordered train-and-check procedure, and execution budgets. Mackey–Glass and Rössler use five independent trajectories; Santa Fe and sunspots use five windows with disjoint test blocks. Repeated circuit settings, measurement samples, and grouping assignments are averaged within each trajectory or window before uncertainty is computed. Every method receives the same raw record and candidate ridge strengths. The three budgets are $(T, N) = (250, 40), (500, 20), (1000, 10)$,

each totaling 10,000 executions. Hardware uses one fixed ten-qubit circuit and one ordered 400-shot record per task and processor, divided into twenty disjoint $N = 20$ blocks. Complete time splits, settings, raw outputs, hardware jobs, checksums, and rebuilding scripts are provided in the supplement and artifact.

REFERENCES

[1] D. Hsu, S. M. Kakade, and T. Zhang, "Random Design Analysis of Ridge Regression," in *Proceedings of the 25th Conference on Learning Theory*, ser. Proceedings of Machine Learning Research, vol. 23, 2012, pp. 9.1–9.24.

[2] W. A. Fuller, *Measurement Error Models*. New York: Wiley, 1987.

[3] K. Fujii and K. Nakajima, "Harnessing Disordered-Ensemble Quantum Dynamics for Machine Learning," *Physical Review Applied*, vol. 8, no. 2, p. 024030, 2017.

[4] K. Nakajima, K. Fujii, M. Negoro, K. Mitarai, and M. Kitagawa, "Boosting Computational Power through Spatial Multiplexing in Quantum Reservoir Computing," *Physical Review Applied*, vol. 11, no. 3, p. 034021, 2019.

[5] P. Mujal, R. Martínez-Peña, J. Nokkala, J. García-Beni, G. L. Giorgi *et al.*, "Opportunities in Quantum Reservoir Computing and Extreme Learning Machines," *Advanced Quantum Technologies*, vol. 4, no. 8, p. 2100027, 2021.

[6] G. Cemin, M. Cech, E. Weiss, S. Soltan, D. Braun *et al.*, "Machine Learning of Reduced Quantum Channels on Noisy Intermediate-Scale Quantum Devices," *Physical Review A*, vol. 110, no. 5, p. 052418, 2024.

[7] C. M. Bishop, "Training with noise is equivalent to Tikhonov regularization," *Neural Computation*, vol. 7, no. 1, pp. 108–116, 1995.

[8] O. Ahmed, F. Tennie, and L. Magri, "Optimal Training of Finitely Sampled Quantum Reservoir Computers for Forecasting of Chaotic Dynamics," *Quantum Machine Intelligence*, vol. 7, p. 31, 2025.

[9] D. A. Kreplin and M. Roth, "Reduction of Finite Sampling Noise in Quantum Neural Networks," *Quantum*, vol. 8, p. 1385, 2024.

[10] C. Moussa, M. H. Gordon, M. Baczyk, M. Cerezo, L. Cincio, and P. J. Coles, "Resource Frugal Optimizer for Quantum Machine Learning," *Quantum Science and Technology*, vol. 8, no. 4, p. 045019, 2023.

[11] F. Hu, G. Angelatos, S. A. Khan, M. Vives, E. Türeci *et al.*, "Tackling Sampling Noise in Physical Systems for Machine Learning Applications: Fundamental Limits and Eigentasks," *Physical Review X*, vol. 13, no. 4, p. 041020, 2023.

[12] T. Kubota, Y. Suzuki, S. Kobayashi, Q. H. Tran, N. Yamamoto, and K. Nakajima, "Quantum Noise-Induced Reservoir Computing," 2022.

[13] A. Palacios de Luis, R. Martínez-Peña, M. C. Soriano, G. L. Giorgi, and R. Zambrini, "Role of Coherence in Many-Body Quantum Reservoir Computing," *Communications Physics*, vol. 7, no. 1, p. 369, 2024.

[14] T. Kubota, Y. Suzuki, S. Kobayashi, Q. H. Tran, N. Yamamoto, and K. Nakajima, "Temporal Information Processing Induced by Quantum Noise," *Physical Review Research*, vol. 5, no. 2, p. 023057, 2023.

[15] R. Martínez-Peña and J.-P. Ortega, "Quantum Reservoir Computing in Finite Dimensions," *Physical Review E*, vol. 107, no. 3, p. 035306, 2023.

[16] P. Mujal, R. Martínez-Peña, G. L. Giorgi, M. C. Soriano, and R. Zambrini, "Time-Series Quantum Reservoir Computing with Weak and Projective Measurements," *npj Quantum Information*, vol. 9, p. 16, 2023.

[17] Y. Suzuki, Q. Gao, K. C. Pradel, K. Yasuoka, and N. Yamamoto, "Natural Quantum Reservoir Computing for Temporal Information Processing," *Scientific Reports*, vol. 12, p. 1353, 2022.

[18] A. Sannia, R. Martínez-Peña, M. C. Soriano, G. L. Giorgi, and R. Zambrini, "Dissipation as a Resource for Quantum Reservoir Computing," *Quantum*, vol. 8, p. 1291, 2024.

[19] L. C. G. Govia, G. J. Ribeill, G. E. Rowlands, and T. A. Ohki, "Nonlinear Input Transformations Are Ubiquitous in Quantum Reservoir Computing," *Neuromorphic Computing and Engineering*, vol. 2, no. 1, p. 014008, 2022.

[20] R. Martínez-Peña, J. Nokkala, G. L. Giorgi, R. Zambrini, and M. C. Soriano, "Information Processing Capacity of Spin-Based Quantum Reservoir Computing Systems," *Cognitive Computation*, vol. 15, no. 5, pp. 1440–1451, 2023.

[21] W. D. Kalfus, G. J. Ribeill, G. E. Rowlands, H. K. Krovi, T. A. Ohki, and L. C. G. Govia, "Hilbert Space as a Computational Resource in Reservoir Computing," *Physical Review Research*, vol. 4, no. 3, p. 033007, 2022.

[22] D. Fry, A. Deshmukh, S. Y.-C. Chen, V. Rastunkov, and V. Markov, "Optimizing Quantum Noise-Induced Reservoir Computing for Nonlinear and Chaotic Time Series Prediction," *Scientific Reports*, vol. 13, p. 19326, 2023.

[23] L. C. G. Govia, G. J. Ribeill, G. E. Rowlands, H. K. Krovi, and T. A. Ohki, "Quantum Reservoir Computing with a Single Nonlinear Oscillator," *Physical Review Research*, vol. 3, no. 1, p. 013077, 2021.

[24] A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, and J. I. Latorre, "Data Re-Uploading for a Universal Quantum Classifier," *Quantum*, vol. 4, p. 226, 2020.

[25] M. C. Mackey and L. Glass, "Oscillation and Chaos in Physiological Control Systems," *Science*, vol. 197, no. 4300, pp. 287–289, 1977.

[26] O. E. Rössler, "An equation for continuous chaos," *Physics Letters A*, vol. 57, no. 5, pp. 397–398, 1976.

[27] N. A. Gershenfeld and A. S. Weigend, "The Future of Time Series," Santa Fe Institute, Tech. Rep. 93-08-053, 1993.

[28] SILSO World Data Center, "Monthly mean total sunspot number V2.0," Royal Observatory of Belgium, Brussels, 2026, dataset SN_m_tot_V2.0, series 1749–2026, accessed July 2026, CC BY-NC 4.0.